

[ARTICLE TITLE]

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Abstract—The Transformer model addresses the limitations of traditional recurrent and convolutional neural networks in sequence modeling, particularly their sequential nature that hinders parallelization and the modeling of long-range dependencies. To overcome these challenges, the Transformer replaces recurrence entirely with a self-attention mechanism, enabling significant parallelization during training and achieving state-of-the-art results in sequence transduction tasks such as machine translation. It employs an encoder-decoder architecture with six identical layers in the base version, each containing multi-head attention mechanisms and position-wise feed-forward networks. Positional information is incorporated via sinusoidal positional encodings, compensating for the absence of recurrence and convolution. Training was conducted on the WMT 2014 English-German and English-French datasets, using byte-pair encoding and word-piece vocabularies, optimized with the Adam algorithm and regularized through label smoothing and dropout. Experiments assessed variations in attention heads, key and value dimensions, and dropout rates, and included application to English syntactic parsing. The large Transformer achieved BLEU scores of 28.4 on English-German and 41.0 on English-French translation tasks, surpassing prior state-of-the-art models by over 2 BLEU points and with markedly lower computational costs. The base model also outperformed other models and ensembles at lower training expense. Additionally, the Transformer attained strong results in English constituency parsing with F1 scores up to 92.7. This architecture enables faster training through parallelization, effectively captures long-range dependencies with reduced computational cost, and shows promise for extension to modalities beyond text, including images, audio, and video, as well as for non-sequential generation tasks.

Index Terms—[KEYWORD 1], [KEYWORD 2], [KEYWORD 3]

I. INTRODUCTION

The field of sequence modeling and transduction has traditionally been dominated by recurrent neural networks (RNNs) and their variants, such as long short-term memory (LSTM) and gated recurrent units (GRU). These models have been widely applied to problems like language modeling and machine translation, due to their ability to process sequences of arbitrary length. However, the inherently sequential nature of RNNs limits parallelization during training and presents difficulties when modeling long-range dependencies within sequences. To address these issues, attention mechanisms have been incorporated into recurrent models, enabling them to capture dependencies between distant positions in input and output sequences more effectively. Despite these advances, attention is commonly used in conjunction with recurrence rather than replacing it.

This work situates itself within this context by exploring a model architecture that entirely removes the recurrent components and relies solely on attention mechanisms to capture

global dependencies. By doing so, it aims to overcome the computational and modeling constraints imposed by recurrent structures, offering both improved parallelization and potentially enhanced performance in sequence transduction tasks.

Bahdanau et al. (2014) [1] introduced an encoder-decoder model with an attention mechanism that addresses the limitation of encoding an entire input sentence into a single fixed-length vector. However, their approach still relies on recurrent neural networks (RNNs) and thus does not remove the inherent sequential computation that restricts training parallelism and efficiency. Similarly, Sutskever et al. (2014) [2] presented deep LSTM models for sequence-to-sequence learning that achieve improved translation performance, but still depend on fixed vector representations and face challenges with long sentences and complex dependencies. Both these influential works do not fully eliminate recurrence, which limits their ability to fully parallelize training and effectively model long-range dependencies. Therefore, there remains a need for a model that completely removes the strict dependency on RNNs and convolutions, employing only attention mechanisms to enable more efficient parallel processing and better handling of global dependencies in sequence transduction.

The central proposal of this work is the Transformer, a novel encoder-decoder architecture that entirely replaces recurrent layers with multi-head self-attention mechanisms and position-wise feed-forward networks to capture global dependencies within sequences. By relying solely on attention, the Transformer eliminates the sequential computation inherent in recurrent architectures, enabling significantly more parallelization during training. Additionally, the model uses sinusoidal positional encodings to inject information about the order of tokens in the sequence, which addresses the absence of recurrence or convolutional structure. This design not only facilitates highly parallel and efficient training but also achieves state-of-the-art performance in sequence transduction tasks like machine translation, directly addressing the limitations found in prior approaches.

- Introduction of the Transformer, the first sequence transduction model based entirely on attention mechanisms without recurrence or convolution (Section 3).
- Implementation of multi-head attention that allows the model to jointly attend to information from different representation subspaces at different positions (Section 3.2).
- Proposal and effective use of sinusoidal positional encodings to provide information about the relative or absolute position of tokens in a sequence (Section 3.5).
- Empirical demonstration that the Transformer outperforms previous state-of-the-art models in English-to-German and English-to-French translation tasks at a fraction of the computational training cost (Section 6.1).

- Additional evaluation showing that the Transformer generalizes well to other tasks such as English constituency parsing (Section 6.3).

The validation of this work is conducted through extensive empirical experiments on large-scale machine translation benchmarks, specifically the WMT 2014 English-to-German and English-to-French datasets. The model is trained using high-performance GPUs to assess its efficiency and scalability. Performance is evaluated on standardized test sets using the BLEU metric, enabling direct comparison with prior state-of-the-art methods such as recurrent and convolutional neural network models and their ensembles. Additionally, the model is tested on other natural language processing tasks, like English constituency parsing, to demonstrate its generalization capabilities. These experiments collectively support the effectiveness and advantages of the proposed Transformer architecture.

The rest of the article is organized into sections that first provide the context and related work, followed by a detailed description of the model architecture including attention mechanisms and feed-forward networks. Subsequent sections cover the datasets and training procedures used, present experimental results and analyses, and conclude with discussions on future research directions.

II. [SECTION 2 TITLE]

[Section 2 content.]

III. [SECTION 3 TITLE]

[Section 3 content.]

IV. [SECTION 4 TITLE]

[Section 4 content.]

V. CONCLUSION

[Conclusions.]

APPENDIX A APPENDIX

[Appendix content.]

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[FIRST AUTHOR] [Brief author biography.]

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